

EX. No. 17 – High-Level Programming Extensions – Function

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MySQL 8.0 Command Line Cli x + v

mysql> CREATE DATABASE IF NOT EXISTS HighLevelDB;
Query OK, 1 row affected, 1 warning (0.03 sec)

mysql> USE HighLevelDB;
Database changed
mysql> DROP TABLE IF EXISTS Employees;
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE Employees (
  -> EmpID INT AUTO_INCREMENT PRIMARY KEY,
  -> EmpName VARCHAR(50),
  -> Salary DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.10 sec)

mysql> INSERT INTO Employees (EmpName, Salary) VALUES
  -> ('Alice', 50000),
  -> ('Bob', 60000),
  -> ('Charlie', 55000);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION SquareNum(n INT)
  -> RETURNS INT
  -> DETERMINISTIC
  -> BEGIN
  -> RETURN n * n;
  -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT SquareNum(5) AS Result;
+-----+
| Result |
+-----+
|    25  |
+-----+
```

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+-----+
|    25  |
+-----+
1 row in set (0.00 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION TotalEmployees()
  -> RETURNS INT
  -> DETERMINISTIC
  -> BEGIN
  -> DECLARE empCount INT;
  -> SELECT COUNT(*) INTO empCount FROM Employees;
  -> RETURN empCount;
  -> END$$
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT TotalEmployees() AS EmployeeCount;
+-----+
| EmployeeCount |
+-----+
|          3    |
+-----+
1 row in set (0.01 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION EmpUpperCase(name VARCHAR(50))
  -> RETURNS VARCHAR(50)
  -> DETERMINISTIC
  -> BEGIN
  -> RETURN UPPER(name);
  -> END$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT EmpName, EmpUpperCase(EmpName) AS UpperName
  -> FROM Employees;
```